

Znaleźć obszar zbieżności szeregów potęgowych

$$\sum_{n=1}^{\infty} \frac{(x+1)^{2n}}{(n+5) \cdot 5^n} = a_n$$

K. DA

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{(x+1)^{2n+2}}{(n+6) \cdot 5^{n+1}} \cdot \frac{(n+5) \cdot 5^n}{(x+1)^{2n}} \right| = \lim_{n \rightarrow \infty} \left| \frac{n+5}{5(n+6)} \right| (x+1)^2 = \frac{1}{5} (x+1)^2 < 1 \text{ zbieżny}$$

$$\sum \frac{n! (x+3)^n}{n^n}$$

K. DA.

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| \rightarrow \left| \frac{(n+1)! (x+3)^{n+1} \cdot n^n}{(n+1)^{n+1} \cdot n! (x+3)^n} \right| \rightarrow$$

$$\rightarrow \left| \frac{\cancel{(n+1)!} \cdot \cancel{(x+3)} \cdot (x+3) \cdot n^n}{(n+1)^n \cdot \cancel{(n+1)} \cdot \cancel{n!} \cdot \cancel{(x+3)^n}} \right| \rightarrow$$

$$\rightarrow \left(\frac{n}{n+1} \right)^n \cdot |x+3| \rightarrow \left(1 + \frac{-1}{n+1} \right)^n \cdot |x+3| \rightarrow$$

$$\rightarrow \left\{ \begin{array}{l} x_1 = \frac{-1}{n+1} \rightarrow 0 \\ y_1 = n \rightarrow \infty \end{array} \right. \quad x_1 y_1 = \frac{-n}{n+1} \rightarrow -1 \quad \rightarrow$$

$$\rightarrow \frac{1}{e} |x+3| < 1$$

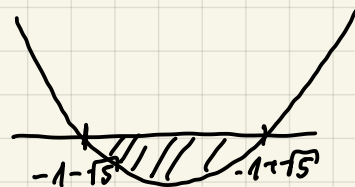
$$|x+3| < e$$

$$e < x+3 < e$$

$$e-3 < x < e-3$$

$$(x+1)^2 < 5$$

$$(x+1-\sqrt{5})(x+1+\sqrt{5}) < 0$$



dla $x \in (-1-\sqrt{5}; -1+\sqrt{5})$ sz. zb.

dla $x \in (-\infty; -1-\sqrt{5}) \cup (-1+\sqrt{5}; +\infty)$ szereg rozst.

co dla $x = -1 \pm \sqrt{5}$ trzeba sprawdzić

$$\sum_{n=1}^{\infty} \frac{(-1+\sqrt{5}+1)^{2n}}{5^n (n+5)} \rightarrow \frac{1}{n+5} \rightarrow \infty$$

$$\sum_{n=1}^{\infty} \frac{(-1-\sqrt{5}+1)^{2n}}{5^n (n+5)} \rightarrow \frac{1}{n+5} \rightarrow \infty$$

Oblicz granice funkcji

$$a) \lim_{x \rightarrow 0} \frac{\sqrt{x+1} - 1}{x} = \left[\frac{0}{0} \right] \rightarrow \frac{\sqrt{x+1} - 1}{x} \cdot \frac{\sqrt{x+1} + 1}{\sqrt{x+1} + 1} \rightarrow$$

$$\rightarrow \frac{\cancel{x}^1 (\sqrt{x+1} + 1)}{\cancel{x}_1 (\sqrt{x+1} + 1)} \rightarrow \frac{1}{2}$$

$$b) \lim_{x \rightarrow 1} \left(\frac{1}{1-x} - \frac{3}{1-x^3} \right) = [\infty - \infty] \rightarrow \left(\frac{1}{(1-x)} - \frac{3}{(1-x)(1+x+x^2)} \right) \rightarrow$$

$$\rightarrow \left(\frac{1+x+x^2-3}{(1-x)(1+x+x^2)} \right) \rightarrow \left(\frac{x^2+x-2}{(1-x)(1+x+x^2)} \right) \rightarrow$$

$$\rightarrow \left(\frac{(x+2)(\cancel{x-1})^1}{(1-\cancel{x})^1 (1+x+x^2)} \right) \rightarrow \frac{-x-2}{1+x+x^2} \rightarrow \frac{-3}{3} \rightarrow -1$$

Wzory na pochodne

$$(x^n)' = n \cdot x^{n-1}$$

$$(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}$$

$$(\sin x)' = \cos x$$

$$(\arccos x)' = \frac{-1}{\sqrt{1-x^2}}$$

$$(\cos x)' = -\sin x$$

$$(\operatorname{arctg} x)' = \frac{1}{1+x^2}$$

$$(\operatorname{tg} x)' = \frac{1}{\cos^2 x}$$

$$(\operatorname{arcc} \operatorname{tg} x)' = \frac{-1}{1+x^2}$$

$$(\operatorname{ctg} x)' = \frac{-1}{\sin^2 x}$$

$$(e^x)' = e^x \quad (\ln x)' = \frac{1}{x}$$

$$f \circ g(x) = f(g(x))$$

$$[f \circ g(x)]' = f'(g(x)) \cdot g'(x)$$

$$(a^x)' = a^x \ln a$$

$$(f \pm g)' = f' \pm g'$$

$$(a \cdot f)' = a \cdot f'$$

$$(f \cdot g)' = f'g + g'f$$

$$\left(\frac{f}{g} \right)' = \frac{f'g - g'f}{g^2}$$

$$f(x) = \ln \ln \ln x$$

$$f'(x) = \frac{1}{\ln \ln x} \cdot (\ln \ln x)' = \frac{1}{\ln \ln x} \cdot \frac{1}{\ln x} \cdot \frac{1}{x}$$

$$f(x) = \arccos x^2$$

$$f'(x) = \frac{-1}{\sqrt{1-x^4}} \cdot (x^2)' = \frac{-1}{\sqrt{1-x^4}} \cdot 2x$$

$$\begin{aligned} [\cos(3x+5) \cdot \cos(3x+5)]' &= \\ &= [-3\sin(3x+5) \cdot \cos(3x+5) - 3] \end{aligned}$$

$$f(x) = (x^2 - 7x + 4)^{25}$$

$$f'(x) = 25(x^2 - 7x + 4)^{24} \cdot (2x - 7)$$

$$f'(x) = \left(\frac{\ln(5x^2 + e^{3x+4})}{\cos^2(3x+5)} \right)' = \frac{\frac{1}{5x^2 + e^{3x+4}} \cdot (10x + e^{3x+4} \cdot 3) \cdot \cos^2(3x+5) - \ln(5x^2 + e^{3x+4}) \cdot (-6\sin(3x+5)\cos(3x+5))}{\cos^4(3x+5)}$$

$$f(x) = x^x = e^{x \ln x}$$

$$f' = e^{g \cdot \ln f} = e^{\ln f^g}$$

$$f'(x) = (e^{x \ln x})' = e^{x \ln x} \left(1 \cdot \ln x + x \cdot \frac{1}{x} \right) = e^{x \ln x} (\ln x + 1)$$

$$f(x) = (\operatorname{arctg} 3x)^{\ln(x^2+2)} = e^{\ln(x^2+2) \cdot \operatorname{arctg} 3x}$$

$$f'(x) = e^{\ln(x^2+2) \cdot \operatorname{arctg} 3x} \left(\frac{1}{x^2+2} \cdot 2x \cdot \operatorname{arctg} 3x + \ln(x^2+2) \cdot \frac{1}{\operatorname{arctg} 3x} \cdot \frac{1}{1+9x^2} \cdot 3 \right)$$

Oblicz pochodną w punkcie x_0 pochodnej funkcji

złożonej wiedząc, że $f(x_0) = 2$, $f'(x_0) = -1$

$$a) f^2(x) = g(x) \quad g'(x) = 2 \cdot f(x) \cdot f'(x)$$

$$g'(x_0) = 2 \cdot f(x_0) \cdot f'(x_0)$$

$$g'(x_0) = 2 \cdot 2 \cdot (-1) = -4$$

$$b) g(x) = \sqrt{1+f(x)} =$$

$$g'(x) = \frac{1}{2\sqrt{1+f(x)}} \cdot f'(x) =$$

$$= \frac{1}{2\sqrt{1+2}} \cdot (-1) = \frac{-1}{2\sqrt{3}} = -\frac{\sqrt{3}}{6}$$

$$c) g(x) = \left(\frac{1-f(x)}{1+f(x)} \right)^2$$

$$g'(x) = 2 \left(\frac{1-f(x)}{1+f(x)} \right) \cdot \left(\frac{1-f(x)}{1+f(x)} \right)' = 2 \left(\frac{1-f(x)}{1+f(x)} \right) \cdot \frac{-f'(x)(1+f(x)) - f(x)(1-f(x))'}{(1+f(x))^2}$$

$$= 2 \left(\frac{(1-f(x)) - 2f(x) \cdot f'(x)}{(1+f(x))^3} \right)$$

$C \rightarrow$ ciągła $D \rightarrow$ różniczkowalna

$f \in D^2(x_p) \Leftrightarrow f, f'$ istnieją i są różniczkowalne na x_p

Twierdzenie (Reguła de l'Hospitala) Reguła H

Służą do bezpośredniego obliczania granic wyrażen' nieoznaczonych typu $\frac{0}{0}$ lub $\frac{\pm\infty}{\pm\infty}$

oraz do pośredniego obliczania granic pozostałych wyrażen' nieoznaczonych.

$(0 \cdot \infty, \infty - \infty, \infty^0, 0^0, 1^\infty)$

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = \left[\frac{0}{0} \right] \stackrel{H}{=} \lim_{x \rightarrow 0} \frac{\cos x}{1} = 1$$

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} \stackrel{H}{=} \lim_{x \rightarrow 0} \frac{(\sin x)'}{x'}$$

$$\lim_{x \rightarrow \infty} \frac{\ln x}{\sqrt{x}} = \left[\frac{\infty}{\infty} \right] \stackrel{H}{=} \lim_{x \rightarrow \infty} \frac{\frac{1}{x}}{\frac{1}{2\sqrt{x}}} = \lim_{x \rightarrow \infty} \frac{2\sqrt{x}}{x} = \lim_{x \rightarrow \infty} \frac{2}{\sqrt{x}} = 0$$

$$\lim_{x \rightarrow 0} \left(\frac{1}{x} - \frac{1}{\sin x} \right) = \lim_{x \rightarrow 0} \frac{\sin x - x}{x \cdot \sin x} = \left[\frac{0}{0} \right] \stackrel{H}{=} \lim_{x \rightarrow 0} \frac{\cos x - 1}{\sin x + x \cos x} = \left[\frac{0}{0} \right] \stackrel{H}{=}$$

$$\stackrel{H}{=} \lim_{x \rightarrow 0} \frac{-\sin x}{\cos x + \cos x - x \sin x} = \left[\frac{0}{2} \right] =$$

$$= 0$$

$$\lim_{x \rightarrow \frac{\pi}{2}^-} (x - \frac{\pi}{2}) \cdot \operatorname{tg} x = [0 \cdot \infty] \rightarrow \frac{x - \frac{\pi}{2}}{\frac{1}{\operatorname{tg} x}} \rightarrow \frac{x - \frac{\pi}{2}}{\operatorname{ctg} x} \rightarrow \left[\frac{0}{0} \right] \stackrel{H}{\rightarrow} \frac{1}{\frac{-1}{\sin^2 x}} \rightarrow -\sin^2 x \rightarrow$$

$$\rightarrow -1$$

$$\lim_{x \rightarrow 0} \left(\frac{1}{x^2} \right)^{\operatorname{tg} x} = [\infty^0] \rightarrow \left\{ f \cdot g = e^{g \cdot \ln f} \right\} \rightarrow e^{\operatorname{tg} x \cdot \ln \frac{1}{x^2}} \rightarrow *$$